

# Admissible Non-Injective Transitions as the Primitive of Physical Description

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## Abstract

Quantum mechanics is the effective theory of physically real unresolved admissible structures and their resolution. This paper establishes the minimal axiomatic foundation from which this claim follows: physical structure is derived from a single primitive, the local structure of admissible transitions between observable states. Four axioms govern this structure: local projective admissibility (A1), structural non-injectivity (A2), admissibility as non-premature selection (A3), and projection locking (A4). From A1 and A2 alone, we derive irreversibility, the arrow of time, and the existence of structural fluctuations, without any additional postulate. From A3, we derive the proto-state as a physically real unresolved configuration shared between two successive states, and show that admissibility forces the retention of phase coherence. We further prove that A3 forbids any admissible factorisation of the fibre, which forces a non-trivial commutator between conjugate generators; combined with minimality (A1) and Born–Infeld parity, this identifies the symmetry group of the admissible fibre as  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  for a prime  $q$ , and the fibre itself as the Weil representation  $V_\rho$  — closing the identification  $F_n \simeq V_\rho$  as a theorem rather than a remark. From A4, we derive the discrete character of quantum transitions as a consequence of the interaction between continuous Born–Infeld saturation and the discrete shell structure of the observable space. This paper provides the axiomatic foundation cited by the companion papers [8, 7, 3] and serves as the formal basis for the derivations carried out therein.

**Keywords.** Non-injective projection; Emergence; Projective structure; Admissibility; Born–Infeld constraint; Information loss; Projection entropy; Quantum emergence; Phase coherence;  $\text{SU}(2)$  structure; Gauge emergence; Fiber structure; Projective temporal ordering; Irreversibility.

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# 1 Introduction

The Cosmochrony programme [2] aims to derive the structure of physical description from a minimal relational foundation, without postulating a background spacetime, a Hilbert space, or an external substrate. The only primitive is the local structure of admissible transitions between observable states.

The central question driving the framework is:

Given an observable state  $O_{n-1}$ , what states are accessible from it?

The answer to this question — encoded in the admissibility structure — is sufficient to derive irreversibility, temporal order, wave-like propagation, phase coherence, and the discrete character of quantum transitions. No substrate, no background, and no quantum postulate is required as input.

## Relation to companion papers

This paper provides the axiomatic foundation for a chain of companion results. The companion paper [7] establishes that Bell-type factorisation fails generically under non-injective projection (A2), without invoking nonlocal dynamics. The companion paper [3] proves that non-injectivity is a structural necessity of any genuinely emergent description. The companion paper [8] derives phase coherence, the singlet correlator, the Tsirelson bound, and the Born rule as structural consequences of admissibility — using Axioms A1–A4 as stated here, combined with the concrete spectral results of the O-series. The white paper [2] develops the full phenomenological programme; the present axiomatic reformulation is a foundational companion to it, not yet integrated into its main text.

## Organisation

Section 2 states the four axioms. Section 3 clarifies the role of  $\Pi$  and the index  $n$ . Section 4 derives the structural consequences of non-injectivity: irreversibility, novelty, and temporal order. Section 5 derives the proto-state, phase, coherence, and the Heisenberg structure of the admissible fibre from the admissibility principle. Section 5.7 derives projection locking and the discrete character of quantum transitions. Section 6 treats entanglement and Bell non-applicability. Section 7 summarises what is derived from what, and identifies the open directions.

## 2 The Four Axioms

We state the four axioms from which the framework proceeds. They are not independent postulates chosen for convenience — each captures a structural feature that is either necessary for genuine emergence [3] or forced by the requirement of bounded effective descriptions [4].

**Definition 2.1** (Admissible transition). *A transition from observable state  $O_{n-1}$  to observable state  $O_n$  is admissible if it satisfies Axioms A1–A4 below.*

**A1 — Local projective admissibility.** For any observable state  $O_{n-1}$ , there exists a non-empty set  $F_n$  of admissible successor directions: configurations of  $O_{n-1}$  compatible with a transition to a stable next state. The set  $F_n$  is defined by internal structural constraints (Born–Infeld bound, spectral envelope, symmetry requirements) and does not presuppose an ambient configuration space.

**A2 — Structural non-injectivity.** Distinct admissible directions from  $O_{n-1}$  may lead to the same resolved successor state  $O_n$ . The map  $\Pi_n$  encoding this is generically non-injective.

**A3 — Admissibility as non-premature selection.** An admissible transition preserves the full multiplicity of open directions until projection locking occurs. It does not select among them prematurely — that is, before the Born–Infeld saturation threshold is reached [4].

**A4 — Projection locking.** Resolution occurs only when continuous Born–Infeld saturation meets the discrete shell support of the observable space. The transition from admissible to admitted is structurally discrete.

**Remark 2.2** (Independence of the axioms). *A1 asserts existence of admissible directions. A2 asserts their generic non-injectivity. A3 constrains which transitions are admissible. A4 governs when resolution occurs. The four axioms are logically independent: none follows from the others. Their joint necessity is argued in Section 3 and in the companion papers [3, 4].*

**Remark 2.3** (Status of Born–Infeld in A3 and A4). *The Born–Infeld saturation threshold appearing in A3 and A4 is not an additional postulate: it is the unique bounded completion of the effective action compatible with bounded projective flux, established in [4]. A3 and A4 therefore do not introduce new structure beyond A1–A2 and the Born–Infeld companion.*

### 3 The Primitive and the Role of $\Pi$

#### 3.1 The primitive

The only primitive of this framework is the pair  $(O_{n-1}, F_n)$ : an observable state and the set of admissible directions it admits. There is no external substrate, no hidden carrier, no background geometry. The fibre  $F_n$  is not defined in an abstract ambient space — it is induced by  $O_{n-1}$  itself through the admissibility constraints of A1–A3.

#### 3.2 The projection $\Pi_n$ as descriptive abstraction

The notation  $\Pi_n$  is a compact encoding of the local admissibility structure. It encodes three things simultaneously:

- (i) the admissible directions from  $O_{n-1}$  — the fibre  $F_n$  (A1, A2);
- (ii) the Born–Infeld saturation bound — the limit on accessible amplitudes (A3);
- (iii) projection locking — the constraint that resolution falls on an available shell (A4).

$\Pi_n$  is not a physical entity and not a dynamical map. It is a structural relation encoding the admissible identification between distinguishable configurations (elements of  $F_n$ ) and indistinguishable observable outcomes (elements of  $O_n$ ). Saying “ $\Pi_n$  is non-injective” means that multiple directions from  $O_{n-1}$  can lead to the same  $O_n$  — a fact about the admissible transition structure, not a property of an external map. Non-injectivity does not originate from an underlying substrate, but from the structure of admissible configurations and their interaction-dependent realisation map.

#### 3.3 The index $n$

The index  $n$  does not represent a pre-existing time parameter. It counts transitions: the number of times a state has been succeeded by another under the admissibility constraints. Its temporal meaning is derived in Section 4 — it is not assumed.

#### 3.4 Where non-injectivity lives

A natural question arises: where does the non-injectivity of  $\Pi_n$  come from, given that no underlying substrate is postulated? The answer is precise.

Non-injectivity is a property of the interaction-dependent realisation map: the map from the space  $F_n$  of structurally compatible configurations to the space  $O_n$  of observable outcomes. When two distinct admissible configurations  $f, f' \in F_n$  give rise to the same observable outcome  $o \in O_n$ , it is not

because they are “the same at a deeper level” — it is because the interaction that realises the transition cannot distinguish them within  $O_n$ . The limit on distinguishability is set by the Born–Infeld capacity constraint (A3): the observable space  $O_n$  has finite projective capacity, and configurations that differ only below this threshold are identified.

Three properties of this realisation map are essential:

- (i) *Interaction-dependence*:  $\Pi_n$  encodes the interaction; different interactions define different realisation maps and different fibres  $F_n$ .
- (ii) *Internal definition of admissibles*:  $F_n$  is defined by internal constraints (BI bound, spectral envelope, symmetry) — not by reference to an ambient substrate.
- (iii) *Minimal structure of  $F_n$* : the admissibility constraints force  $F_n$  to carry a specific algebraic structure —  $SU(2)$  in the  $\text{spin-}\frac{1}{2}$  sector (O23) — which is derived from the constraints, not assumed.

The non-injectivity is therefore neither a property of a hidden substrate nor an arbitrary assumption: it is a structural consequence of finite projective capacity applied to an internally defined space of admissible configurations.

**Remark 3.1** (Connection to the O-series:  $F_n$  and  $V_\rho$ ). *In the concrete spectral realisation of the O-series, the admissible fibre  $F_n$  is realised as a spectral sector  $V_\rho$  of the Heisenberg Cayley graph  $\text{Cay}(\text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$ . The sectors  $V_\rho$  are not pre-existing configuration spaces: they are the structures induced by the admissibility constraints — the Born–Infeld bound, the spectral envelope, and the symmetry requirements of A1. In this sense,  $V_\rho$  is the concrete realisation of  $F_n$ , not an ambient space in which  $F_n$  is embedded:*

$$F_n \simeq V_\rho. \tag{1}$$

*This identification, stated here as a structural connection to the O-series, is established as a theorem in Section 5.4: the axioms A1–A3 and Born–Infeld parity together force the symmetry group of  $F_n$  to be  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ , and the Stone–von Neumann theorem then identifies  $F_n \simeq V_\rho$  uniquely.*

**Remark 3.2** (The two levels of  $\Pi_n$ : kernel and image (O24)). *The non-injective projection  $\Pi_n$  operates at two structurally distinct levels, established in O24 [11]:*

- (i) *Kernel level ( $\ker \Pi_n$ ): governs microscopic multiplicity and fluctuation amplitude. Multiple configurations in  $\Omega_n^{(c)}$  may project to the same observable outcome; their differences are invisible to admissible observables and appear as projection residuals. The dependence on  $O_{n-1}$  — via the baseline  $V_{n-1}^{(c)}$  — enters at this level: it determines which directions in  $\Omega_n^{(c)}$  are new relative to the already realized subspace.*

- (ii) Image level ( $\text{Im } \Pi_n \cap \text{Ntrl}$ ): governs the observable structure. Its real rank is rigidly fixed to 3 by the quaternionic maximality of  $\text{Im } \mathbb{H}$  (O23, Theorem 3.2) and is independent of the cardinality of  $\ker \Pi_n$ .

The Verticality Lemma of O24 (Lemma 3.1) establishes that every Born–Infeld-admissible symmetry acts vertically: it maps each fibre  $\Pi_n^{-1}(o)$  to itself and does not increase the rank of the image. The structural consequence is:

Non-injectivity affects degeneracy, not structure.

Enlarging  $\ker \Pi_n$  — admitting more configurations in each fibre — increases the amplitude of quantum fluctuations but does not alter the observable rank, the cascade exponent  $\delta_{\text{pair}}$ , or any other structural invariant of the admissible sector. The two levels of  $\Pi_n$  are therefore decoupled: kernel structure is physically relevant for fluctuations; image structure is physically relevant for observables.

The abstract structure of Section 3 has a concrete realisation in the O-series that answers precisely where  $\Omega_n$  lives and how it is constructed. The realisation proceeds through five levels, each induced by the previous one.

**Level 1 — The relational group  $G_q$ .** The combinatorial substrate of the BFS exploration is the discrete Heisenberg group

$$G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), \quad |G_q| = q^3, \quad (2)$$

equipped with the standard symmetric generating set  $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$ .  $G_q$  is not a physical space: it is a relational structure, encoding how configurations relate to one another via the generators. The BFS explores  $G_q$  in shells  $S_n = B_n \setminus B_{n-1}$ , where  $B_n = \{g \in G_q : d(e, g) \leq n\}$ . The shell structure  $\{S_n\}$  is the concrete realisation of the transition-counting index  $n$  of Section 3.

**Level 2 — The Weil representation  $\rho_c$  and its ambient space.** For each non-zero character  $c \in (\mathbb{Z}/q\mathbb{Z})^\times$ , the discrete Heisenberg group  $G_q$  admits an irreducible  $q$ -dimensional Weil representation [5]:

$$\rho_c : G_q \longrightarrow \text{GL}(\mathbb{C}^q), \quad (\rho_c(a, b, \gamma) f)(x) = e^{2\pi i c(\gamma + bx)/q} f(x - a), \quad (3)$$

acting on  $L^2(\mathbb{Z}/q\mathbb{Z}) \cong \mathbb{C}^q$ . The ambient space  $\mathbb{C}^q$  is not a physical configuration space presupposed by the theory: it is the natural carrier of the irreducible representation of  $G_q$  of central character  $c$ . It is induced by  $G_q$  and  $c$ , not by any external assumption.

**Level 3 — The orbit span  $\Omega_n$  as emerging subspace.** Given the initial vector  $v_0 \in \mathbb{C}^q$  and the Weil action  $\rho_c$ , the orbit span at BFS depth  $n$  is:

$$\Omega_n^{(c)} := \text{span}\{\rho_c(g)v_0 : g \in B_n\} \subseteq \mathbb{C}^q. \quad (4)$$

By construction,  $\Omega_n^{(c)}$  is a minimal extension of the previously realized subspace:

$$V_{n-1}^{(c)} \subseteq \Omega_n^{(c)} \subseteq \mathbb{C}^q, \quad (5)$$

where  $V_{n-1}^{(c)} = \text{GS}(B_{n-1})$  is the Gram–Schmidt span accumulated over the preceding shells.  $\Omega_n^{(c)}$  is not defined by  $O_{n-1}$ , but is constrained by the already realized subspace  $V_{n-1}^{(c)}$ : the directions in  $\Omega_n^{(c)} \setminus V_{n-1}^{(c)}$  are the candidates for new observable structure at depth  $n$ .

**Definition 3.3** (Structurally compatible continuations).  $\Omega_n^{(c)}$  is the space of structurally compatible continuations from  $O_{n-1}$ : it contains every direction reachable by the relational structure of  $G_q$  via  $\rho_c$ , without any admissibility filter applied. Admissibility is not a property of  $\Omega_n^{(c)}$ ; it is a property of  $\Pi_n$  acting on  $\Omega_n^{(c)}$ .

Three properties follow immediately:

- (i) *Emergence*:  $\Omega_n^{(c)}$  is not given a priori — it is built incrementally as the BFS explores  $G_q$ .
- (ii) *Indirect conditioning on  $O_{n-1}$* :  $\Omega_n^{(c)}$  is defined by the group action alone — the definition (4) refers only to  $B_n$  and  $\rho_c$ . The dependence on  $O_{n-1}$  is indirect:  $O_{n-1}$  determines the Gram–Schmidt span  $\text{GS}_{n-1}$  already accumulated over  $B_{n-1}$ , which acts as the baseline against which new directions in  $\Omega_n^{(c)}$  are measured. This baseline is used by  $\Pi_n$  (not by  $\Omega_n^{(c)}$  itself) to identify genuinely new directions, counted by  $\sigma_c(n) = \delta r_n / |S_n|$  [5]. In summary:  $\Omega_n^{(c)}$  is determined by  $(G_q, \rho_c, B_n)$ ; the role of  $O_{n-1}$  enters only through  $\Pi_n$ .
- (iii) *Pre-admissibility character*:  $\Omega_n^{(c)}$  contains all directions generated by the group action of  $G_q$  up to depth  $n$  — including directions that  $\Pi_n$  will subsequently reject as inadmissible. The admissibility filter is entirely in  $\Pi_n$ , not in  $\Omega_n^{(c)}$ .

**Level 4 — The admissibility filter  $\Pi_n$ .** The projection  $\Pi_n$  acts on  $\Omega_n^{(c)}$  via the BI admissibility constraint  $A_n \leq c_\chi / \sqrt{\lambda_n}$  and the Gram–Schmidt orthogonalisation [5, 10]. It selects, from all directions in  $\Omega_n^{(c)}$ , those compatible with the admissibility constraints.  $\Pi_n$  encodes the laws revealed by the O-series — Born–Infeld bound, parity involution, projection locking — but does not define  $\Omega_n^{(c)}$ .



**Level 5 — The observable outcome  $O_n$ .** The observable state  $O_n$  is the image of  $\Pi_n$  applied to the admissible part of  $\Omega_n^{(c)}$ . It is a discrete, shell-indexed element of the observable space [10]:

$$O = \bigsqcup_{n \in \mathbb{N}_q} O_n. \quad (6)$$

**Remark 3.4** (Summary: the five-level hierarchy). *The five levels constitute a single construction with no external inputs:*

$$G_q \xrightarrow{\rho_c} \mathbb{C}^q \xrightarrow{\text{BFS}} \Omega_n^{(c)} \xrightarrow{\Pi_n} O_n, \quad V_{n-1}^{(c)} \subseteq \Omega_n^{(c)} \subseteq \mathbb{C}^q. \quad (7)$$

$G_q$  is the relational structure.  $\mathbb{C}^q$  is its representation carrier.  $\Omega_n^{(c)}$  is the emergent space of structurally compatible continuations (not yet filtered by admissibility), constrained from below by the already realized subspace  $V_{n-1}^{(c)}$ .  $\Pi_n$  is the admissibility filter.  $O_n$  is the resolved observable state. No level presupposes the next: each is induced by the preceding structure and the admissibility constraints. This is the concrete answer to the question of where  $\Omega_n$  lives: in  $\mathbb{C}^q$ , itself induced by  $G_q$  — not in an external configuration space.

**Remark 3.5** (The conjugate pair and  $\Omega_n$  without preferred direction). The parity involution  $c \leftrightarrow q - c$  (O18) acts on  $\Omega_n^{(c)}$  via complex conjugation [6]:  $\Omega_n^{(q-c)} = \overline{\Omega_n^{(c)}}$ . The minimal admissible fibre is therefore the pair  $\{V_n^{(c)}, V_n^{(q-c)}\} = \{V_n^{(c)}, \overline{V_n^{(c)}}\}$ , with no preferred orientation — consistent with the admissibility principle of A3.

### 3.5 From $\mathbb{C}^q$ to continuous spacetime: the large- $q$ limit

The five-level hierarchy of Section 3.4 is formulated for finite prime  $q$ . Physical spacetime is continuous. This subsection identifies the structural bridge between the two, without claiming to complete the derivation.

**The discrete structure at finite  $q$ .** At finite  $q$ , the observable space  $O = \bigsqcup_{n \in \mathbb{N}_q} O_n$  is discrete and finite. The ambient representation space  $\mathbb{C}^q$  has dimension  $q$ . Physical quantities are functions on  $\mathbb{Z}/q\mathbb{Z}$ .

**The large- $q$  limit.** As  $q \rightarrow \infty$  with the prime structure retained, three things happen simultaneously:

- (i) *The representation space  $\mathbb{C}^q$  approaches  $L^2(\mathbb{R})$ .* Functions on  $\mathbb{Z}/q\mathbb{Z}$  with fixed Fourier support become functions on  $\mathbb{R}$  in the Pontryagin dual limit. The Weil representation  $\rho_c$  of  $G_q$  approaches the metaplectic representation of the real Heisenberg group  $\text{Heis}_3(\mathbb{R})$ .

- (ii) *The BFS shell structure approaches a continuous foliation.* The polynomial ball growth  $|B_n| \sim c n^4$  (Bass–Guivarc’h, homogeneous dimension  $D = 4$ ) induces, in the  $q \rightarrow \infty$  limit, a continuous layering of the configuration space compatible with a four-dimensional effective geometry [2].
- (iii) *The discrete admissibility filter  $\Pi_n$  approaches a differential constraint.* The Born–Infeld admissibility bound  $A_n \leq c_\chi / \sqrt{\lambda_n}$  becomes, in the continuum limit, the Born–Infeld Lagrangian constraint on effective field configurations [4].

The continuum spacetime observable is therefore not a separate postulate: it is the large- $q$  limit of the discrete observable space  $O$ , with the four-dimensional structure inherited from the homogeneous dimension of  $G_q$ .

**Remark 3.6** (Status of the continuum limit). *The convergence  $\mathbb{C}^q \rightarrow L^2(\mathbb{R})$  and the emergence of continuous spacetime geometry from the BFS shell structure are established qualitatively in [2]. A fully rigorous derivation of the Lorentzian metric from the spectral structure of  $G_q$  in the  $q \rightarrow \infty$  limit is the subject of the companion spectral geometry programme and remains open at the level of an explicit theorem. The present remark identifies the mechanism — not the proof.*

**Remark 3.7** (Why four dimensions). *The homogeneous dimension  $D = 4$  of the Heisenberg group  $G_q$  (Bass–Guivarc’h theorem:  $|B_n| \sim n^4$ ) is not chosen to match the observed four dimensions of spacetime. It is a structural consequence of the nilpotency class and rank of  $\text{Heis}_3$ : the group has three generators with one non-trivial commutator, giving  $D = 2 \cdot \text{rank} + \text{class} = 4$ . The coincidence with the observed dimension of spacetime is therefore a structural prediction of the framework, not a parameter fit.*

## 4 Structural Consequences of Non-Injectivity

Non-injectivity (A2) and the existence of distinct successor states together imply three structural facts whose sources are distinct.

### 4.1 Irreversibility

**Proposition 4.1** (Irreversibility from non-injectivity). *Under A2, every admissible transition involves an irreversible loss of structural information.*

*Proof.* Since multiple directions from  $O_{n-1}$  map to the same  $O_n$  under  $\Pi_n$ , the information distinguishing these directions is absent from  $O_n$ . No subsequent transition  $\Pi_{n'}$  for  $n' > n$  can recover this information: the map  $\Pi_n$  has positive projection entropy  $S_{\Pi_n} > 0$  [3], and no admissible transition can reduce  $S_{\Pi}$  below zero. The loss is therefore structural and permanent.  $\square$

### 4.2 Novelty and structural fluctuations

The existence of a distinct successor state  $O_n \neq O_{n-1}$  defines a structural residue:

$$\Delta_n := \text{Res}(O_{n-1}, O_n; \Pi_n), \quad (8)$$

the measure of how much  $O_n$  is irreducible to a continuation of  $O_{n-1}$  under  $\Pi_n$ . That  $\Delta_n$  is non-trivial is not an additional assumption — it is the condition under which we index  $O_n$  as a distinct state. These residues are what physical descriptions call fluctuations or quantum noise.

### 4.3 Temporal order

**Theorem 4.2** (Temporal order from A2). *Under A2, the chain of observable states  $\{O_n\}_{n \geq 0}$  carries a canonical oriented, non-trivial partial order.*

*Proof.* Irreversibility (Proposition 4.1) establishes an asymmetric relation:  $O_{n-1} \rightarrow O_n$  is not reversible, so  $O_{n-1}$  and  $O_n$  are not equivalent under the transition structure. Novelty ( $\Delta_n \neq 0$ ) ensures  $O_n \neq O_{n-1}$ , so the order is non-trivial. The composition of transitions defines transitivity. Together: asymmetry, irreflexivity, and transitivity give a strict partial order compatible with the chain structure.  $\square$

**Remark 4.3** (Time as derived structure). *The arrow of time is not postulated. It is the oriented order generated by the chain of non-injective transitions under A2. The index  $n$  acquires its temporal meaning from this order — not the reverse.*

## 5 Admissibility, Wave, and Phase Coherence

### 5.1 The admissibility principle

Axiom A3 introduces a physical constraint on transitions:

A transition is admissible if and only if it preserves the full multiplicity of open admissible directions without selecting among them prematurely.

Premature selection means making a determinative choice that nothing in the current state  $O_{n-1}$  justifies. Admissibility forbids it: the transition must remain structurally neutral among all open directions until projection locking (A4) forces resolution.

### 5.2 The proto-state

**Definition 5.1** (Proto-state). *The proto-state  $\Psi_n$  associated with a transition from  $O_{n-1}$  to  $O_n$  is the unresolved configuration shared between  $O_{n-1}$  and  $O_n$ : the carrier of the transition, belonging fully to neither endpoint.*

The proto-state is physically real — it is the configuration of the system during the transition, before projection locking. It is not an epistemic representation of uncertainty about a definite state: it is the actual structural state of the system between two realized observable states.

**Proposition 5.2** (Structural indeterminacy of the proto-state). *Under A1 and A3, the proto-state  $\Psi_n$  is genuinely indeterminate among the admissible directions in  $F_n$ .*

*Proof.* By A1,  $F_n$  is non-empty. By A3, the transition does not select prematurely among the elements of  $F_n$ . Therefore  $\Psi_n$  is not identified with any particular element of  $F_n$  before projection locking: it is open across all of  $F_n$  simultaneously. This is structural indeterminacy, not epistemic ignorance.  $\square$

### 5.3 Wave, phase, and coherence

The proto-state, being structurally open across all elements of  $F_n$  simultaneously, propagates through all admissible directions at once. This is the physical origin of wave-like behaviour: not a metaphor, but a direct consequence of admissibility.

A configuration propagating through multiple directions simultaneously carries relative phase structure between those directions. Phase is therefore not postulated — it is a consequence of structural indeterminacy.

Admissibility (A3) preserves this phase structure: premature selection would collapse the relative phases, but A3 forbids it. The chain of consequences is therefore:

$$\begin{aligned}
\text{A3} &\implies \text{structural indeterminacy} \\
&\implies \text{proto-state (wave)} \\
&\implies \text{phase} \\
&\implies \text{coherence preserved.}
\end{aligned} \tag{9}$$

**Remark 5.3** (The admissible fibre as precursor of Hilbert structure). *The fibre  $F_n$ , equipped with the phase structure forced by A3, is the natural precursor of an effective Hilbert sector: it carries complex amplitudes and interference structure. This is not yet a full Hilbert space — the inner product, completeness, and probability measure require additional derivation (see Q1, Q2 in Section 7). But the amplitude structure and interference follow from A3 alone.*

**Remark 5.4** (Double-slit experiment). *In the double-slit experiment, the proto-state propagates through both slits because the admissibility constraints do not select between them: both are open directions in  $F_n$ . The interference pattern is a direct consequence of coherent propagation through all admissible directions. The impact on the screen is the moment the proto-state reaches the Born–Infeld saturation threshold — projection locking (A4) forces it onto an available shell, resolving it into a localized state  $O_n$ .*

## 5.4 Non-factorisability and the emergence of Heisenberg structure

Axiom A3 does not only enforce phase coherence. It also constrains the algebraic structure of the admissible fibre  $F_n$  itself, forcing it to carry a Heisenberg structure rather than an arbitrary unitary symmetry.

**Lemma 5.5** (Admissible non-factorisability). *Under A1–A3, any decomposition*

$$F_n = F^{(1)} \oplus F^{(2)}$$

*that is stable under the admissible generators of  $F_n$  is forbidden prior to projection locking. In particular, A3 forces the action of admissible generators on  $F_n$  to be irreducible.*

*Proof.* A3 forbids premature selection among open directions. A decomposition of  $F_n$  stable under admissible generators separates directions by invariants that are internal to the fibre — invariants carried by the generators themselves, which are constitutive of the admissibility structure by A1. Such a decomposition therefore resolves the proto-state partially, by assigning each open direction to a definite sector, prior to any Born–Infeld

saturation event. This contradicts A3. Irreducibility follows immediately: any proper stable sub-representation would yield such a forbidden decomposition.  $\square$

**Theorem 5.6** (A1–A3 and BI parity force Heisenberg structure). *Under A1–A3 and Born–Infeld parity, the symmetry group of the admissible fibre  $F_n$  is isomorphic to  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  for some prime  $q$ . Consequently,  $F_n$  is isomorphic to the Weil representation  $V_\rho$  on  $L^2(\mathbb{Z}/q\mathbb{Z})$ .*

*Proof sketch.* Let  $X$  and  $\sigma(X)$  be the conjugate generators of  $F_n$  under Born–Infeld parity. If  $[X, \sigma(X)] = 0$ , then  $F_n$  decomposes into simultaneous eigenspaces, yielding a stable decomposition forbidden by Lemma 5.5. Hence  $[X, \sigma(X)] \neq 0$ . Minimality (A1) forces the commutator  $Z := [X, \sigma(X)]$  to be central: any non-central  $Z$  would extend the algebra beyond the conjugate pair, contradicting the minimal generative closure forced by A1. The group  $\langle X, \sigma(X) \rangle$  is then a non-trivial central extension of  $(\mathbb{Z}/q\mathbb{Z})^2$  by  $\langle Z \rangle$ , isomorphic to  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  by the classification of such extensions. Lemma 5.5 further implies that  $|Z|$  is prime: if  $|Z|$  were composite, the representation of  $F_n$  would decompose over the characters of  $\langle Z \rangle$ , yielding a forbidden stable decomposition. Finally, Born–Infeld parity forces  $c \neq 0$  [4], so the central character is non-trivial; the discrete Stone–von Neumann theorem then identifies  $F_n \simeq V_\rho$  uniquely. The detailed proof of each step is given in the companion paper [1].  $\square$

$$\begin{aligned}
& \text{A3} \implies \text{no admissible factorisation} \implies [X, \sigma(X)] \neq 0 \implies \text{central extension,} \\
& \text{A1} \implies \text{minimal generative closure} \implies \text{Heisenberg algebra,} \\
& \text{A3} \implies \text{irreducibility} \implies |Z| = q \text{ prime} \implies \text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), \\
& \text{BI parity} \implies c \neq 0 \implies F_n \simeq V_\rho.
\end{aligned} \tag{10}$$

**Remark 5.7** (Uncertainty principle as a corollary). *Theorem 5.6 subsumes the open direction Q6 of Section 7. The incompatibility of conjugate observables  $X$  and  $\sigma(X)$  — i.e. the Heisenberg uncertainty principle — is a direct corollary: it follows from  $[X, \sigma(X)] \neq 0$  together with the standard operator inequality. The uncertainty principle is therefore not a property of the observables, but a property of the non-resolved proto-state: it is the algebraic signature of the prohibition on premature selection encoded in A3.*

**Remark 5.8** (Canonical quantisation as structural necessity). *The relation  $[X, \sigma(X)] = Z$  defines a minimal discrete symplectic structure, of which the Heisenberg algebra is the canonical quantisation. Theorem 5.6 therefore establishes that canonical quantisation is not a modelling choice: it is the unique algebraic structure compatible with the requirement that the proto-state remain genuinely non-resolved until projection locking.*

A quantum superposition is an unresolved proto-state: a physically real configuration whose fibre maintains coherent relations between multiple open admissible directions. There is no ontological paradox — superposition is the normal condition of a system between two transitions.

A measurement is a transition whose admissibility constraints are strong enough to force resolution of the proto-state onto a single admissible direction. Free evolution and measurement differ in the strength of their admissibility constraints, not in their fundamental mechanism.

In this sense, quantum mechanics is the effective theory of physically real unresolved admissible structures and their resolution.

### 5.5 The wavefunction as effective representation of the proto-state

A quantum wave is not a physical substance propagating in space. It is the effective representation of a physically real unresolved proto-state. It encodes the coherent structure of admissible directions prior to projection locking.

More precisely: the proto-state  $\Psi_n$  is real; the wavefunction  $\psi$  is a coordinate description of the coherent admissible structure carried by  $\Psi_n$  in the spectral realisation  $V_\rho$  of  $F_n$ . When we write  $\psi$ , we are not describing a substance propagating in space — we are encoding the relational phase structure of an unresolved configuration in the space of admissible directions. Quantum mechanics does not describe realized physical states: it describes states in the process of being resolved.

This resolves the standard interpretational difficulties without additional postulate:

- *Superposition* is the normal condition of a proto-state — the simultaneous structural openness across all elements of  $F_n$ .
- *Interference* is the direct consequence of coherent propagation through multiple admissible directions before resolution.
- *Collapse* is not a physical process applied to a pre-existing state but the resolution of the proto-state at projection locking — the transition from unresolved admissible structure to a realized observable outcome.

### 5.6 Why complex amplitudes

Given that the proto-state is a physically real, unresolved, coherent configuration, what is the minimal mathematical structure capable of representing it?

Real positive weights are insufficient. They describe ignorance about which of several pre-existing alternatives is realized — a classical mixture.

The proto-state is not a mixture: it is a single coherent configuration with definite phase relations between admissible directions.

Real signed weights are also insufficient. They allow opposition of sign but do not support a rich enough phase structure under composition. In particular, they cannot represent the full coherence of a proto-state whose directions combine under the  $SU(2)$  structure forced by O23.

The minimal structure that can: (i) represent simultaneous coherent openness across multiple directions, (ii) encode their relative phase relations, (iii) compose correctly under the  $SU(2)$  symmetry of the admissible fibre, (iv) support interference — the physical consequence of coherence before resolution — is a complex amplitude.

Therefore: complex amplitudes are not introduced by convention. They are the minimal representation of a coherent unresolved proto-state. The imaginary unit encodes the relative phase between admissible directions; it enters because the proto-state is not a mixture but a coherent superposition.

## 5.7 Why the Born weight is the squared norm

Upon projection locking (A4), the unresolved proto-state is resolved onto a single admissible direction. The quantity associated with each direction at resolution — the *intensity of realisation* — must satisfy four structural requirements:

- (i) *Positivity*: a weight on admissible directions cannot be negative.
- (ii) *Global phase invariance*: multiplying  $\psi$  by a global phase  $e^{i\theta}$  does not change the physical proto-state, so the weight must be invariant under this transformation.
- (iii) *Normalisability*: the total weight over all admissible directions must be finite and summable to one.
- (iv) *Coherence preservation before resolution*: the weight must not erase phase relations prematurely — interference must remain possible up to the moment of resolution.

The squared norm  $|\psi|^2$  is the unique minimal quantity satisfying all four requirements. The norm  $|\psi|$  fails (ii) under phase shifts and (iv) — it introduces real-valued interference which is not the correct structure. Any higher power  $|\psi|^p$  for  $p \neq 2$  fails (iii) under composition of superpositions. Any non-analytic function of  $\psi$  fails (iv).

The Born weight is therefore not a postulate. It is the positive intensity associated with the resolution of a coherent admissible structure — the unique minimal such intensity compatible with the four structural requirements.



**Remark 5.9** (Relation to the Born rule derivation in the companion paper). *The argument above establishes the structural necessity of the squared norm as the resolution intensity of the proto-state. The companion paper [8] makes this precise as a theorem in the  $SU(2)$  parity sector: the Born rule  $P(a = +1|\hat{a}) = |\langle +\hat{a}|\psi\rangle|^2$  is the unique probability assignment compatible with the coherent fibre structure, the  $SU(2)$  observable algebra, and the derived singlet correlator. The two derivations are complementary: the present argument gives the structural motivation; the companion gives the uniqueness proof.*

**Remark 5.10** (The wavefunction lives in the admissible fibre, not in space). *A common source of confusion is to picture the wavefunction as an object propagating in physical space. In the present framework,  $\psi$  encodes the structure of  $F_n$  — the admissible directions from  $O_{n-1}$ . Its concrete realisation is the spectral sector  $V_\rho$  induced by the admissibility constraints on the Heisenberg Cayley graph ( $O$ -series).  $F_n$  and  $V_\rho$  are not pre-existing spaces: they are the spaces defined by the admissibility constraints. The wavefunction does not propagate in space; it encodes the relational phase structure of an unresolved admissible configuration.*

Axiom A4 establishes that resolution is discrete. This discreteness is not postulated: it follows from the interaction of two structures already present in the framework.

## 5.8 Two types of discreteness

The Born–Infeld saturation condition [4] operates on a continuous amplitude function. Its saturation locus  $\mathcal{L}_{\text{BI}} \subset \mathbb{R}$  is generically real-valued — its elements carry no information about shell structure.

The observable space decomposes discretely:

$$O = \bigsqcup_{n \in \mathbb{N}_q} O_n, \quad (11)$$

indexed by BFS shells of the Heisenberg Cayley graph [10]. No component  $O_{n'}$  exists for non-admissible depths.

The proto-state is resolved at the first element of  $\mathcal{L}_{\text{BI}} \cap \mathbb{N}_q$  — the first depth where Born–Infeld saturation coincides with an available shell. This is the Projection Locking theorem [10].

**Theorem 5.11** (Discreteness from projection locking, after O22). *Under A3 and A4, quantum transitions are discrete. The discreteness is a consequence of the interaction of continuous Born–Infeld saturation with the discrete admissibility filter of  $\Pi_n$ , not a postulate.*

### 5.9 Three stable directions

The number of independent stable admissible directions in the fibre is exactly three, derived algebraically from the admissibility constraints alone [12]:

$$\begin{aligned}
\text{BI parity} &\Rightarrow \text{fibre involution } c \leftrightarrow q - c \\
&\Rightarrow \text{non-abelian admissible support} \\
&\Rightarrow \mathbb{H}\text{-minimality} \\
&\Rightarrow \dim_{\mathbb{R}} \text{Im } \mathbb{H} = 3 \\
&\Rightarrow \Sigma_c(n_3) = 3.
\end{aligned} \tag{12}$$

The emergence of exactly three stable directions — and with it, three particle generations — is algebraic in origin, not geometric or phenomenological.

## 6 Entanglement and Bell Non-Applicability

### 6.1 Entanglement as shared ancestral fibre

Two observable systems  $A$  and  $B$  are entangled when they share a common ancestral fibre: a past state  $O_{n-k}$  in which they formed a single non-factorisable configuration.

The shared fibre  $F_{n-k}$  reduces the admissible directions for each subsystem: the admissible directions for  $A$  are constrained by  $B$ , and vice versa. The correlations arise not from any interaction at the moment of measurement, but from the persistence of the shared fibre — a residue of the past transition in which  $A$  and  $B$  were joined.

### 6.2 Why Bell’s factorisation assumptions do not apply

Bell’s theorem [13] constrains models in which measurement outcomes are determined by a variable  $\lambda$  satisfying: (i)  $\lambda$  is local (assignable separately to  $A$  and  $B$ ), (ii)  $\lambda$  is present (defined at the moment of measurement), (iii)  $\lambda$  is independent (statistically independent of measurement settings).

The fibre variable in the present framework satisfies none of these conditions. It is not local (it belongs to the shared fibre, not to a spatial region), not present (it belongs to a past transition, not the current state), and not independent ( $A$  and  $B$  share the same ancestral fibre, which cannot be factorised).

The Bell factorisation assumption  $P(a, b|\lambda) = P(a|\lambda) \cdot P(b|\lambda)$  fails because it requires  $\lambda$  to be assignable separately to  $A$  and  $B$ : the shared fibre is structurally non-factorisable. Bell-type inequalities are therefore not violated in this framework — their preconditions do not hold. This is established formally in the companion paper [7].

## 7 Summary: What Is Derived from What

Table 1 summarises the derivation chain.

| Concept                                    | Source                                              | Status              |
|--------------------------------------------|-----------------------------------------------------|---------------------|
| Irreversibility                            | Non-injectivity (A2)                                | Proved (Prop. 4.1)  |
| Structural fluctuations                    | Relational residue $\Delta_n$                       | Derived             |
| Temporal order                             | Irreversibility + novelty (A2)                      | Proved (Thm. 4.2)   |
| Proto-state (wave)                         | Unresolved multiplicity (A1, A3)                    | Derived (Prop. 5.2) |
| Structural indeterminacy                   | Multiple open directions (A1, A3)                   | Derived             |
| Phase coherence                            | Admissibility preserves open directions (A3)        | Derived             |
| Coherent amplitude structure               | Admissible fibre + phase                            | Derived             |
| Interference                               | Coherent amplitude structure                        | Derived             |
| Discrete transitions                       | Projection locking (A4, O22)                        | Proved (Thm. 5.11)  |
| Three stable directions                    | $\dim_{\mathbb{R}} \text{Im } \mathbb{H} = 3$ (O23) | Proved              |
| Superposition                              | Unresolved proto-state                              | Derived             |
| Collapse                                   | Proto-state resolution at locking                   | Derived             |
| Entanglement                               | Shared non-factorisable ancestral fibre             | Derived             |
| Bell non-applicability                     | Temporal, non-local, non-independent fibre          | Proved [7]          |
| Phase coherence $\Rightarrow$ QM           | Admissibility + O-series                            | Proved [8]          |
| Born rule (SU(2) sector)                   | Correlator + coherence (Q1)                         | Proved [8]          |
| $F_n \simeq V_\rho$ (Heisenberg structure) | Non-factorisability (A1, A3) + BI parity            | Proved (Thm. 5.6)   |
| Uncertainty principle                      | $[X, \sigma(X)] \neq 0$ (Thm. 5.6)                  | Proved (Rem. 5.7)   |
| Full Hilbert structure                     | Beyond SU(2) parity sector                          | <b>Open</b>         |
| Born rule (general)                        | Beyond SU(2) parity sector                          | <b>Open</b>         |

Table 1: Derivation chain from the four axioms. “Derived” denotes a consequence that follows from the axioms by structural argument; “Proved” denotes a result with an explicit proof either in this paper or in a cited companion.

### Open directions

**Q2 — Born rule beyond SU(2).** The Born rule  $P(a = +1|\hat{a}) = |\langle +\hat{a}|\psi \rangle|^2$  is derived in [8] for the SU(2) parity sector. A first step beyond this sector is taken in [9], which derives the Born rule, singlet correlator, and Tsirelson bound for the co-admissible spin- $\frac{3}{2}$  sector on the binary icosahedral group  $2I$ , and identifies SU(2) as the unique stable sector under admissibility refinement in the large- $p$  limit. The full extension to arbitrary observables, multipartite systems, and continuous spectra requires extending the fibre structure beyond the two-element parity case and remains the primary open direction.

**Q5 — Proto-state ontology.** The proto-state is physically real and shared between  $O_{n-1}$  and  $O_n$ . Theorem 5.6 closes the primary open ques-

tion of this direction: the space in which the proto-state exists between transitions is  $F_n \simeq V_\rho$ , the Weil representation of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ , determined uniquely by A1–A3 and Born–Infeld parity. The relationship between this discrete structure and continuous spacetime geometry in the large- $q$  limit (Section 3.5) remains the subject of the companion spectral geometry programme.

**Q6 — Uncertainty principle as fibre incompatibility.** This direction is resolved by Theorem 5.6 and Remark 5.7: the Heisenberg uncertainty principle follows as a corollary of  $[X, \sigma(X)] \neq 0$ , which is itself a consequence of A3 alone. The uncertainty principle is a structural property of the non-resolved proto-state, not an independent postulate about observables.

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